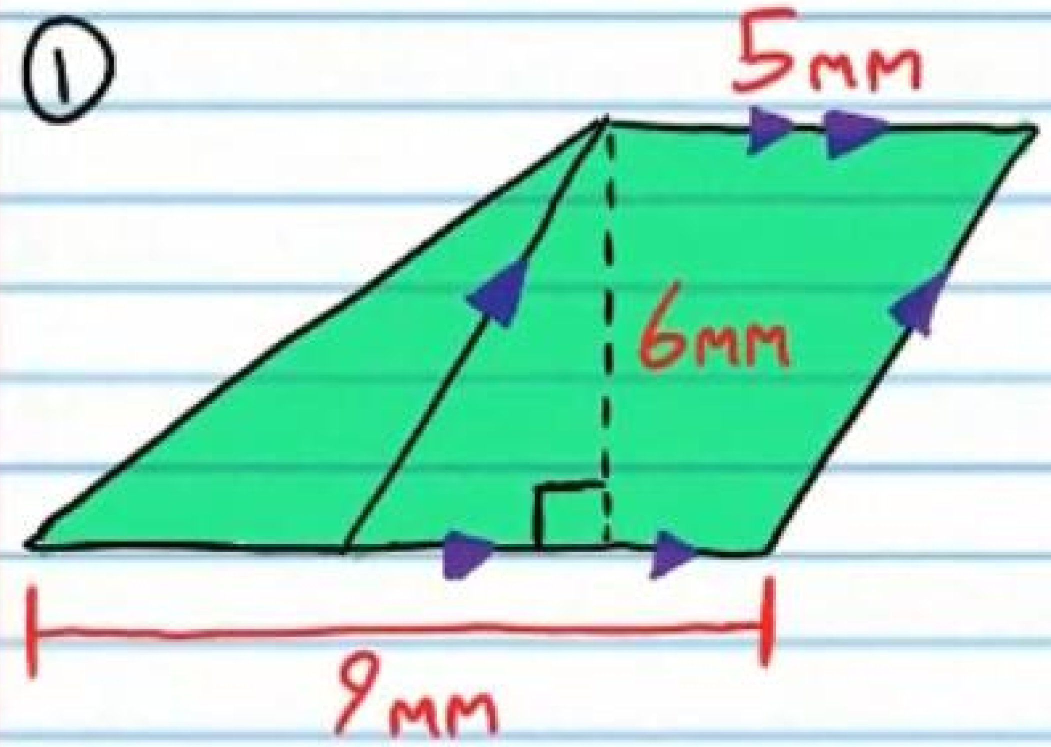
Practice Geometry Test VI (Version I)

Name
Date
Course

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show all work.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
V) Each problem is worth four points for a total of 100 points.

Find the area of each shaded region below.

① 

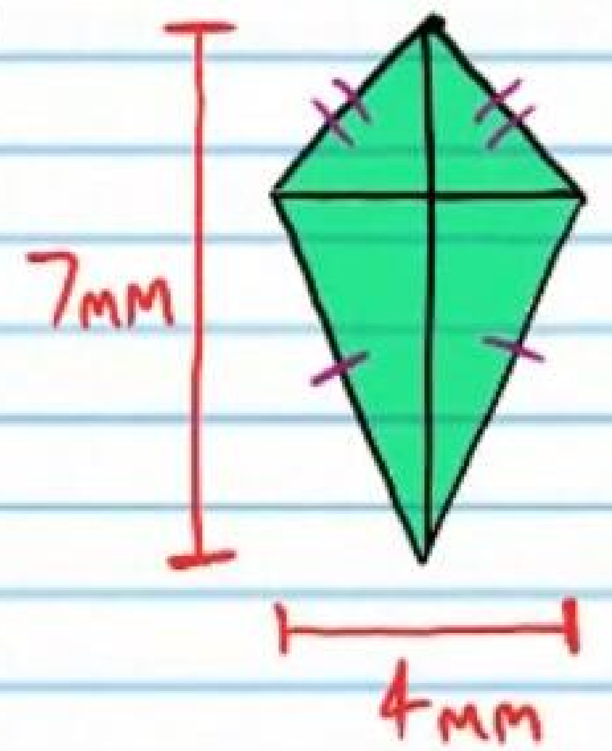
$A = A_{par} + A_{tri}$

$A = 5(6) + \frac{1}{2}(9-5)(6)$

$A = 30 + \frac{1}{2}(4)6$

$A = 30 + 12$

$A = \boxed{42\ mm^2}$

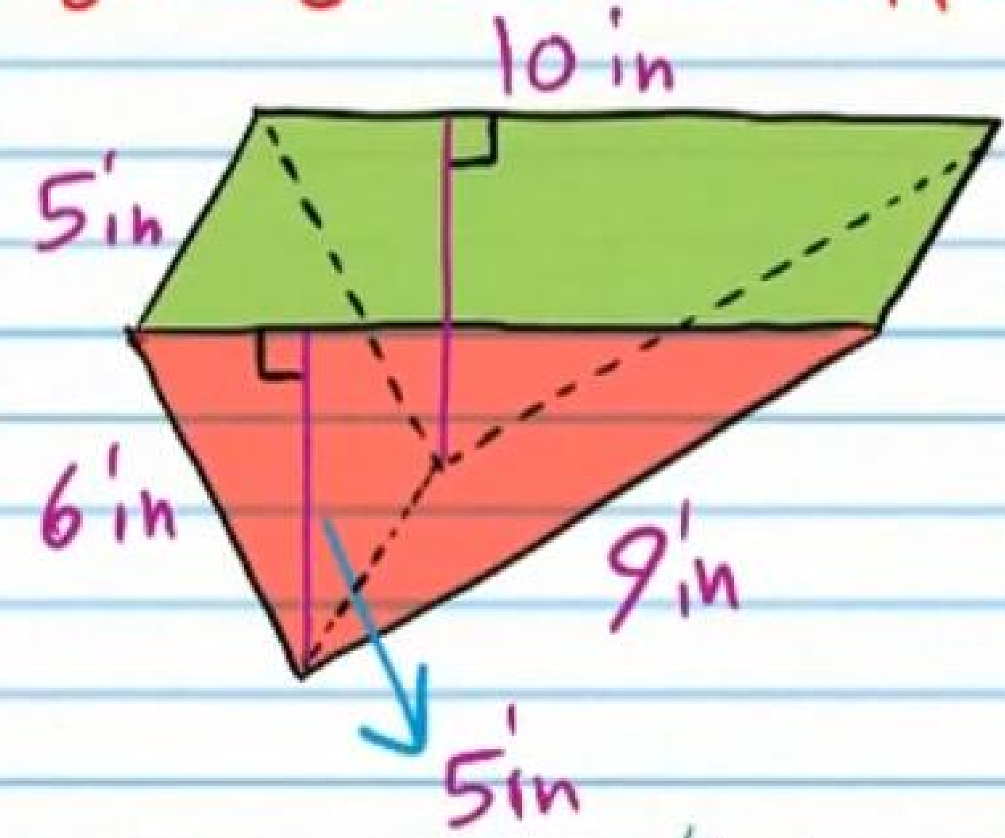
② 

$A = \frac{1}{2} D_1 D_2$

$A = \frac{1}{2}(7)(4)$

$A = \boxed{14\ mm^2}$

③ Find the surface area of the prism below. Assume right angles where apparent.

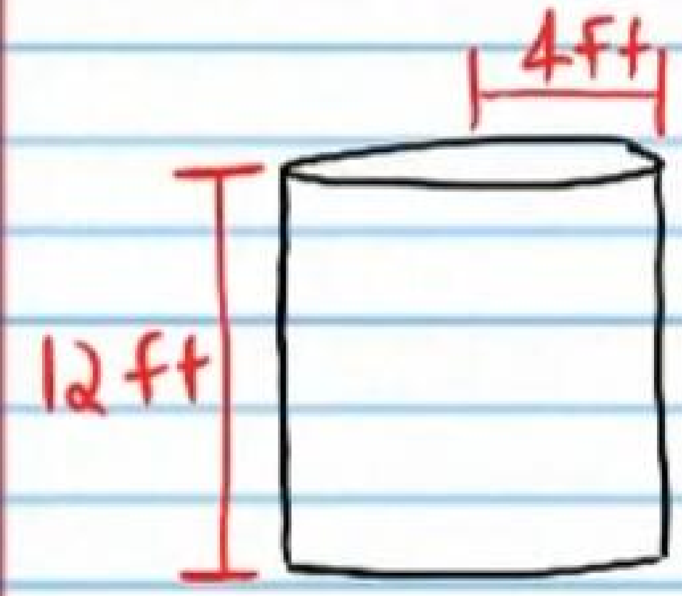


$A = 5(10) + 2\left(\frac{1}{2}(10)(5)\right) + 5(6) + 9(5)$

$A = 50 + 50 + 30 + 45$

$A = \boxed{175\ in^2}$

④ Find the surface area of the right cylinder below. Leave your answer as a multiple of pi or round your answer to the nearest tenth.



$A = 2\pi r^2 + 2\pi r h$

$A = 2\pi (4)^2 + 2\pi(4)(12)$

$A = 2\pi (16) + 96\pi$

$A = 32\pi + 96\pi$

$A = \boxed{128\pi\ ft^2}$

Use $\odot M$ below to answer problem #'s 5-7.

⑤ Find the area of sector BMC. Round your answer to the nearest ten thousandth.

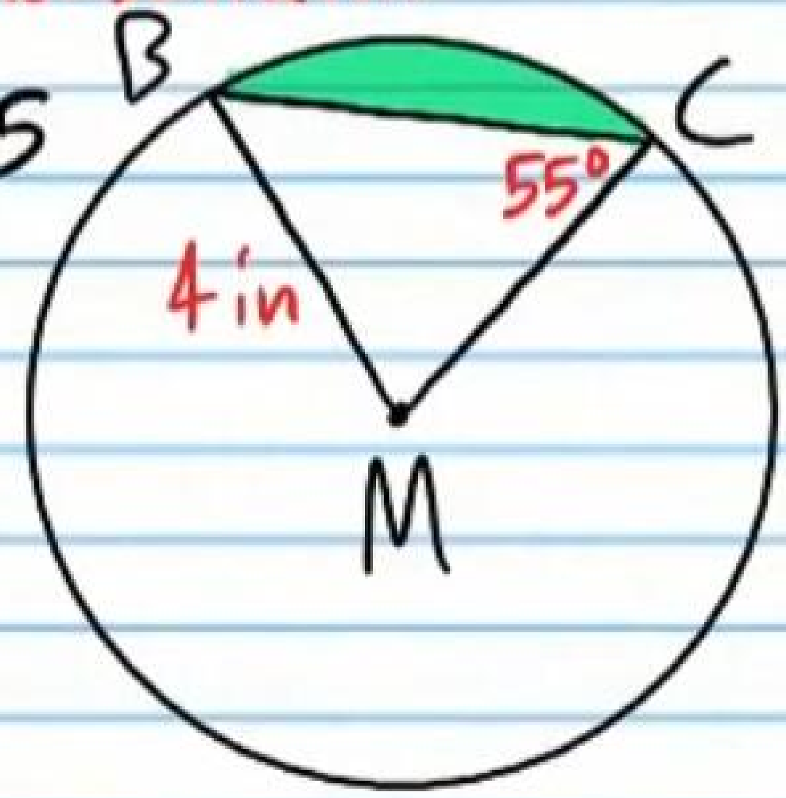
$A = \frac{\theta}{360}\pi r^2$

$A = \frac{70}{360}\pi(4)^2$

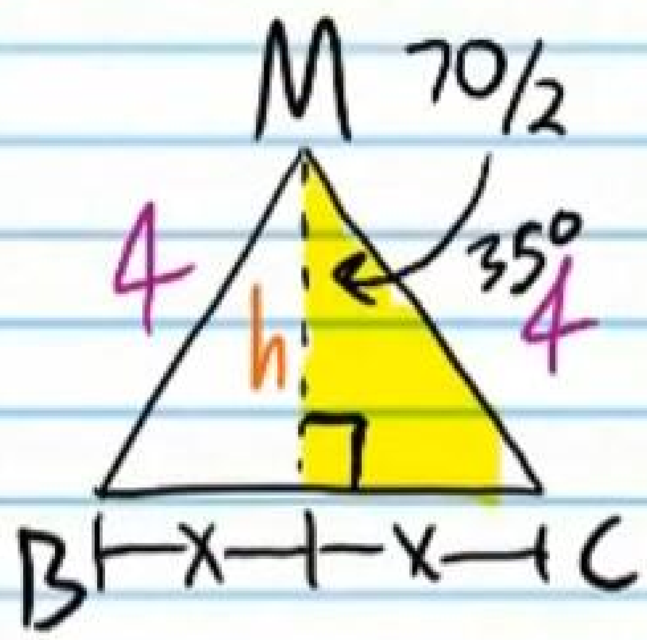
$A = \boxed{9.7738 \text{ in}^2}$

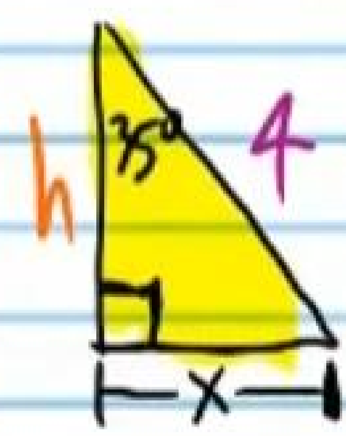
$m\angle BMC = 180 - 55 - 55$

$= 70°$



⑥ Find the area of $\triangle BMC$. Round your answer to the nearest ten thousandth.





$\cos 35° = \frac{h}{4}$

$4\cos 35° = h$

$\sin 35° = \frac{x}{4}$

$4\sin 35° = x$

$b = 2x = 2(4\sin 35°) = 8\sin 35°$

$A = \frac{1}{2}bh = \frac{1}{2}(8\sin 35°)(4\cos 35°)$

$= 16\sin 35°\cos 35°$

$= \boxed{7.5175 \text{ in}^2}$

⑦ Find the area of the segment (i.e. the shaded region). Round your answer to the nearest tenth.

$A_{seg} = A_{sect} - A_{Tri} \approx 9.7738 - 7.5175 \approx \boxed{2.3 \text{ in}^2}$

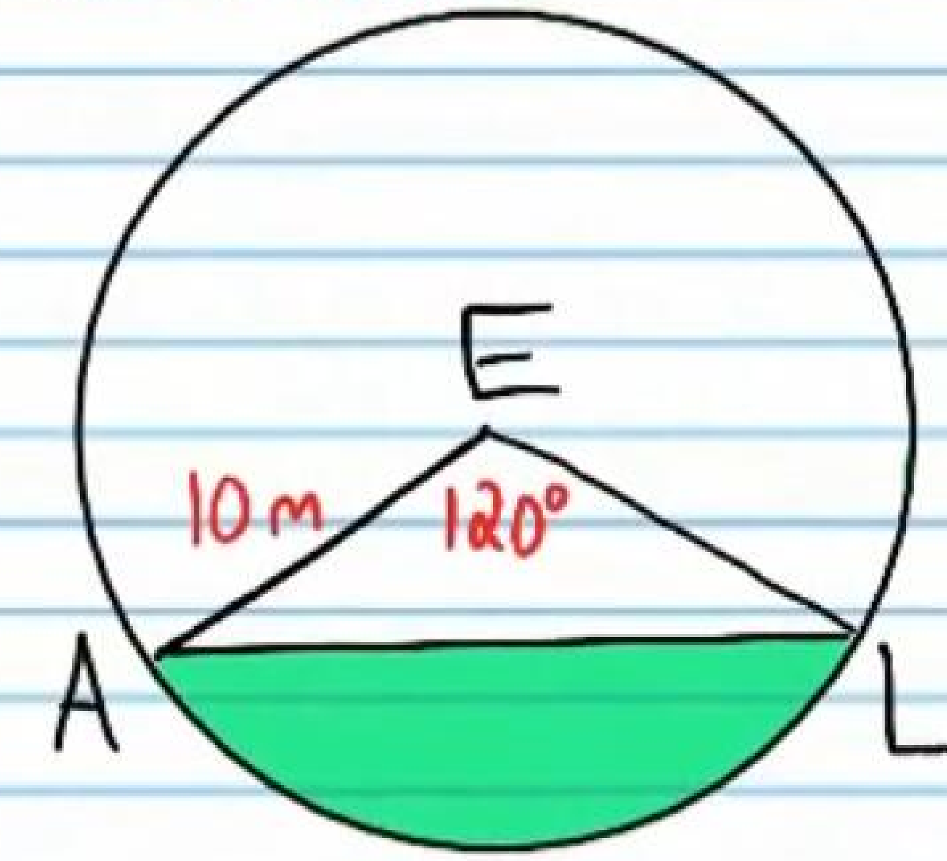
Use $\odot E$ below to answer problem #'s 8–10.

⑧ Find the area of sector AEL. Round your answer to the nearest ten thousandth.

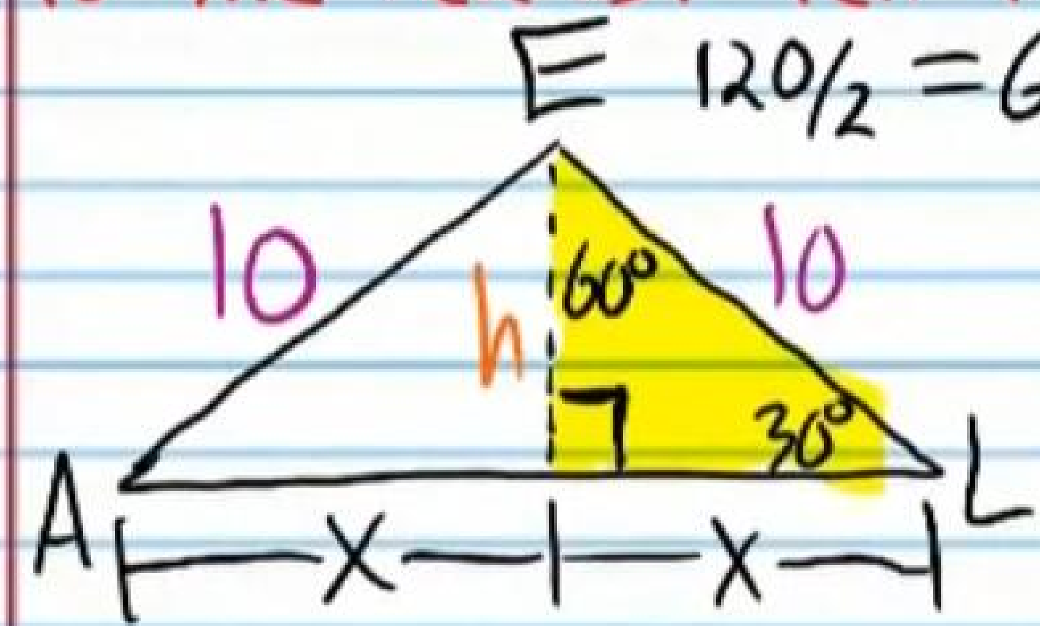
$A = \frac{\theta}{360}\pi r^2$

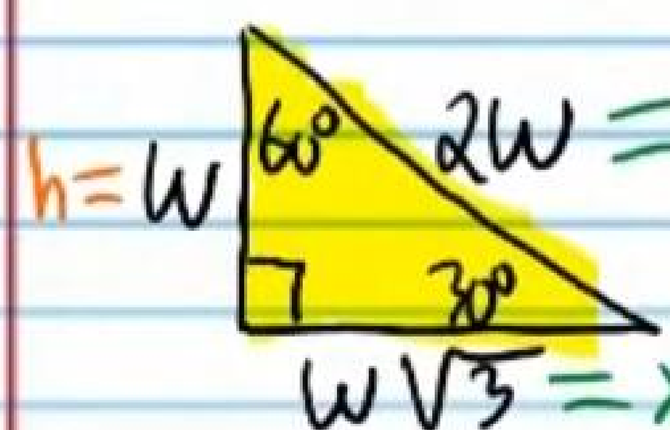
$A = \frac{120}{360}\pi(10)^2$

$A \approx \boxed{104.7198 \text{ m}^2}$



⑨ Find the area of $\triangle AEL$. Round your answer to the nearest ten thousandth.





$2w = 10$

$w = 5$

$h = 5,\ b = 2x = 2(5\sqrt{3})$

$= 10\sqrt{3}$

$A = \frac{1}{2}bh$

$A = \frac{1}{2}(10\sqrt{3})(5)$

$A = 25\sqrt{3}$

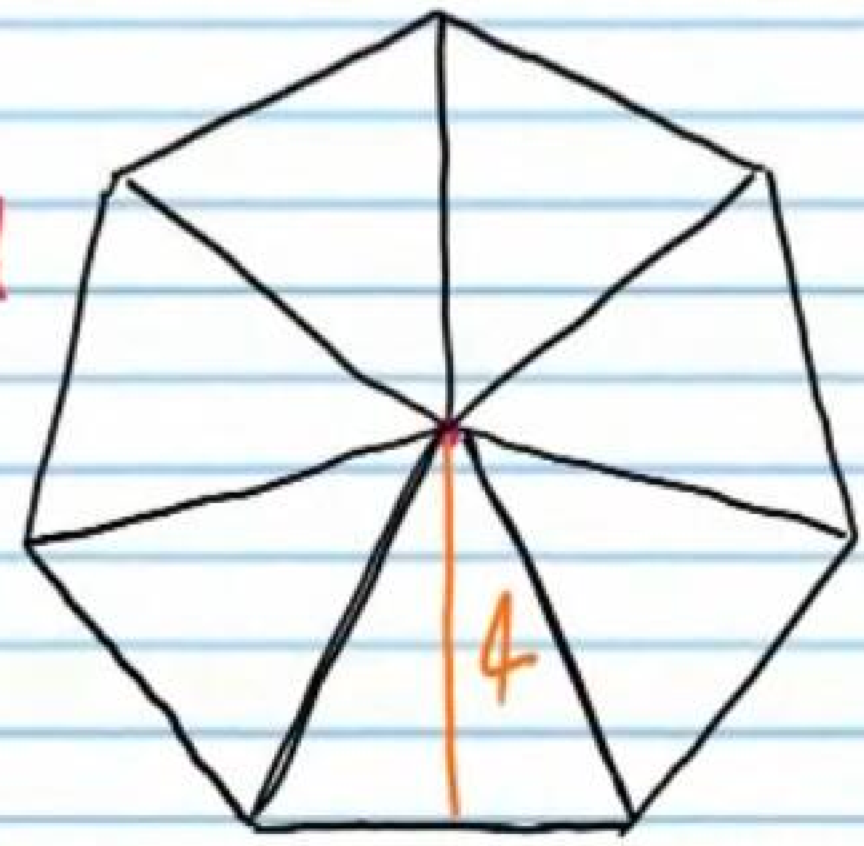
$A = \boxed{43.3013 \text{ m}^2}$

⑩ Find the area of the segment (i.e. the shaded region). Round your answer to the nearest tenth.

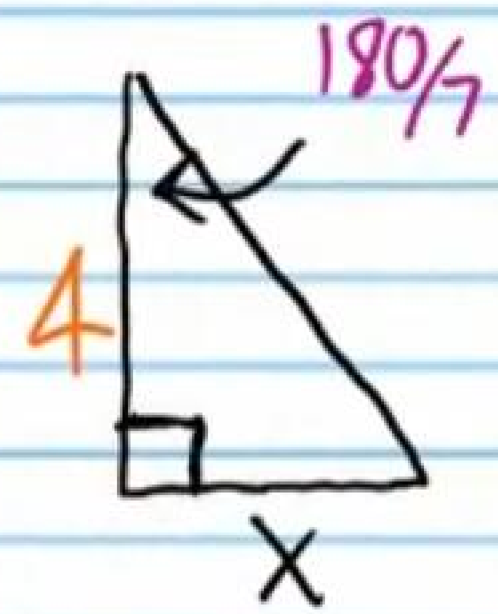
$$A_{seg} = A_{sect} - A_{Tri} \approx 104.7198 - 43.3013 \approx \boxed{61.4\ m^2}$$

Use the regular polygon below to answer problem #'s 11-12.

⑪ Find the side length if the apothem is 4 feet. Round your answer to the nearest ten thousandth.



$$\frac{360}{7} \div 2 = \frac{360}{14} = \frac{180}{7}$$



$$\tan\left(\frac{180}{7}\right)^\circ = \frac{x}{4}$$

$$4\tan\left(\frac{180}{7}\right)^\circ = x$$

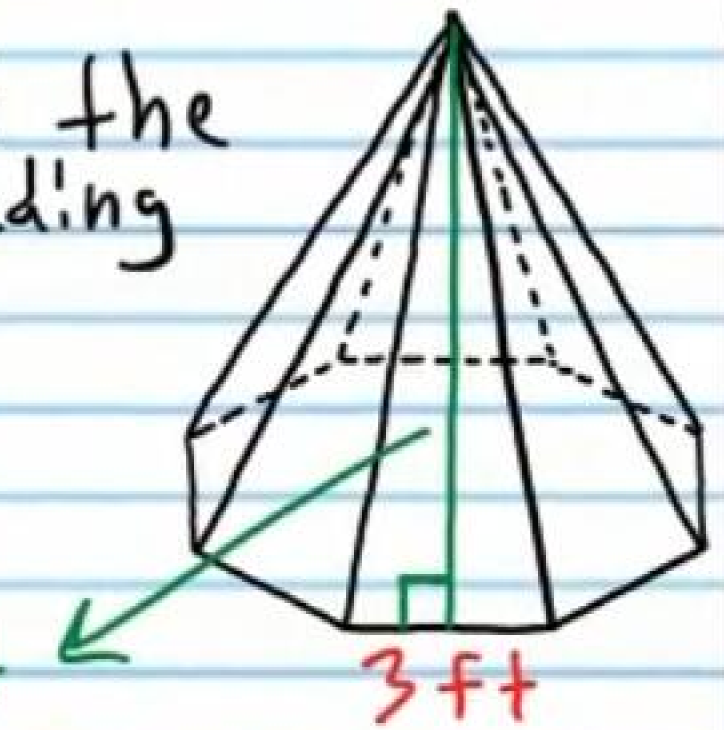
$$b = \text{side length} = 2x = 2\left(4\tan\left(\frac{180}{7}\right)^\circ\right)$$
$$= 8\tan\left(\frac{180}{7}\right)$$
$$\approx \boxed{3.8526\ ft}$$

⑫ Find the area of the polygon. Round your answer to the nearest tenth.

$$A = 7A_{Tri} = 7\left(\tfrac{1}{2}bh\right) \approx 7\left(\tfrac{1}{2}(3.8526)(4)\right) \approx \boxed{53.9\ ft^2}$$

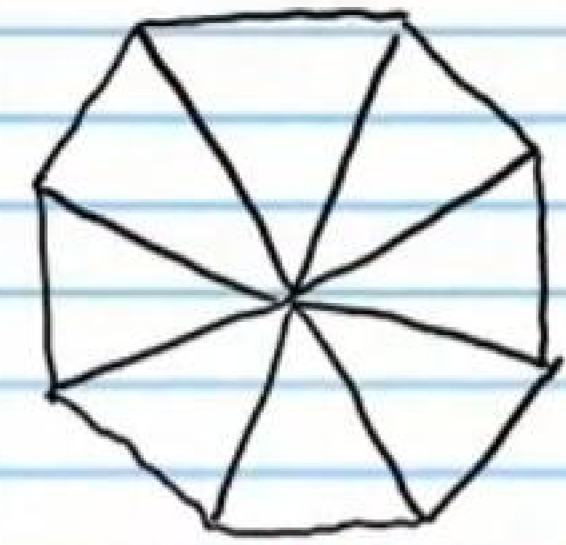
Use the regular pyramid below to answer problem #'s 13-15.

⑬ Find the total area of all of the sides of the pyramid (not including the base).



$$A = 8A_{Tri} = 8\left(\tfrac{1}{2}(3)(7)\right) = \boxed{84\ ft^2}$$

⑭ Find the area of the base of the pyramid. Round your answer to the nearest ten thousandth.



3ft

$\frac{360}{8} = 45^\circ$

$\frac{45}{2} = 22.5^\circ$

$3/2 = 1.5$

$$\tan 22.5^\circ = \frac{1.5}{h}$$

$$h = \frac{1.5}{\tan 22.5^\circ}$$

$$A = 8A_{Tri} = 8\left(\tfrac{1}{2}(3)\left(\frac{1.5}{\tan 22.5^\circ}\right)\right)$$
$$= \frac{18}{\tan 22.5^\circ}$$
$$\approx \boxed{43.4558\ ft^2}$$

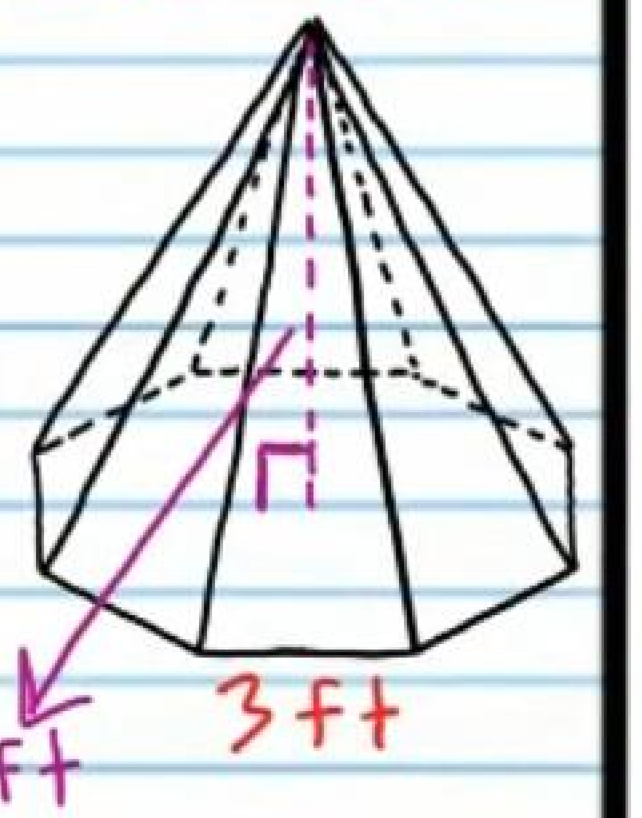
(15) What is the total surface area of the pyramid? Round your answer to the nearest tenth.

$$A = A_{sides} + A_{Base} \approx 84 + 43.4558 \approx \boxed{127.5\ ft^2}$$

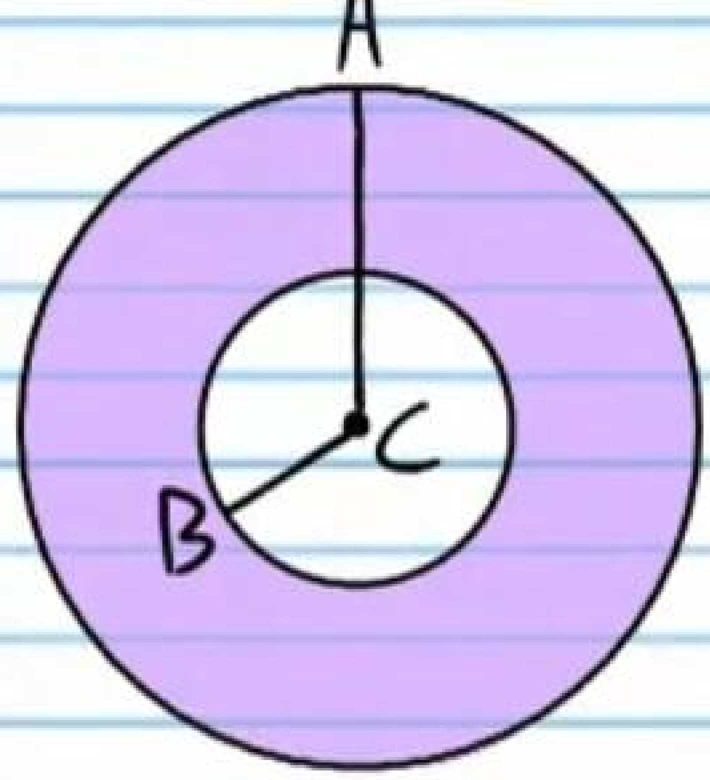
(16) Find the volume of the pyramid in problem #'s 13-15 if its height is 6 feet. Round to the nearest tenth.

$$V = \frac{1}{3}Bh = \frac{1}{3}(43.4558)(6)$$

$$= \boxed{86.9\ ft^3}$$



(17) Find the probability that a randomly chosen point in the figure will lie in the shaded region. if $AC = 7m$ and $BC = 3m$.



$$P = \frac{A_{shaded}}{A_{Total}}$$

$$= \frac{\pi(7)^2 - \pi(3)^2}{\pi(7)^2}$$

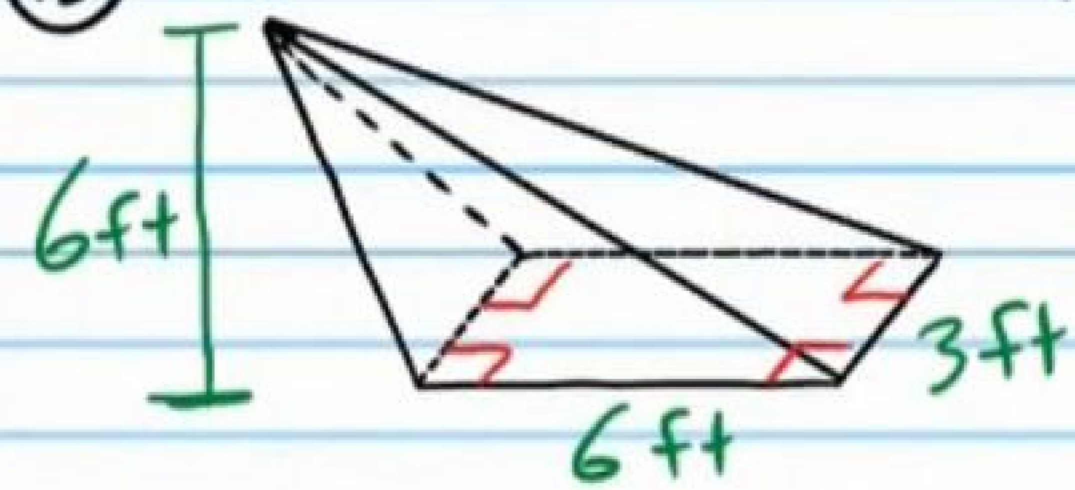
$$= \frac{49\pi - 9\pi}{49\pi}$$

$$= \frac{40\pi}{49\pi} = \boxed{\frac{40}{49}}$$

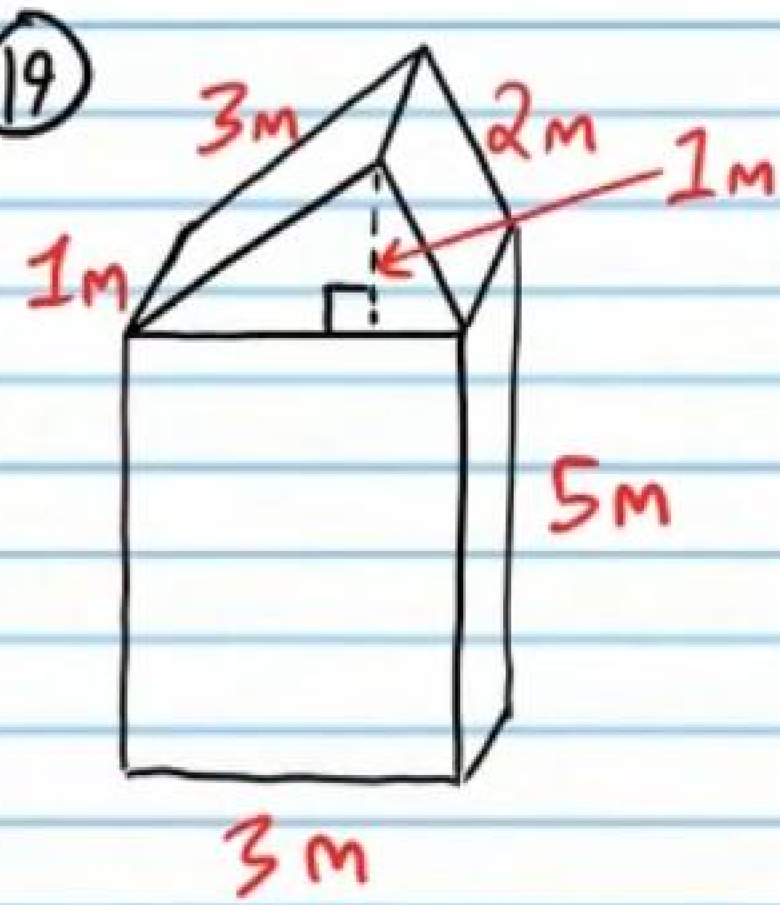
Find the volume of the following figures. Assume right angles where apparent and round to the nearest tenth.

(18)

$$V = \frac{1}{3}Bh = \frac{1}{3}(6 \cdot 3)(6) = \boxed{36\ ft^3}$$



(19)



$$V = V_{Rect} + V_{Tri}$$

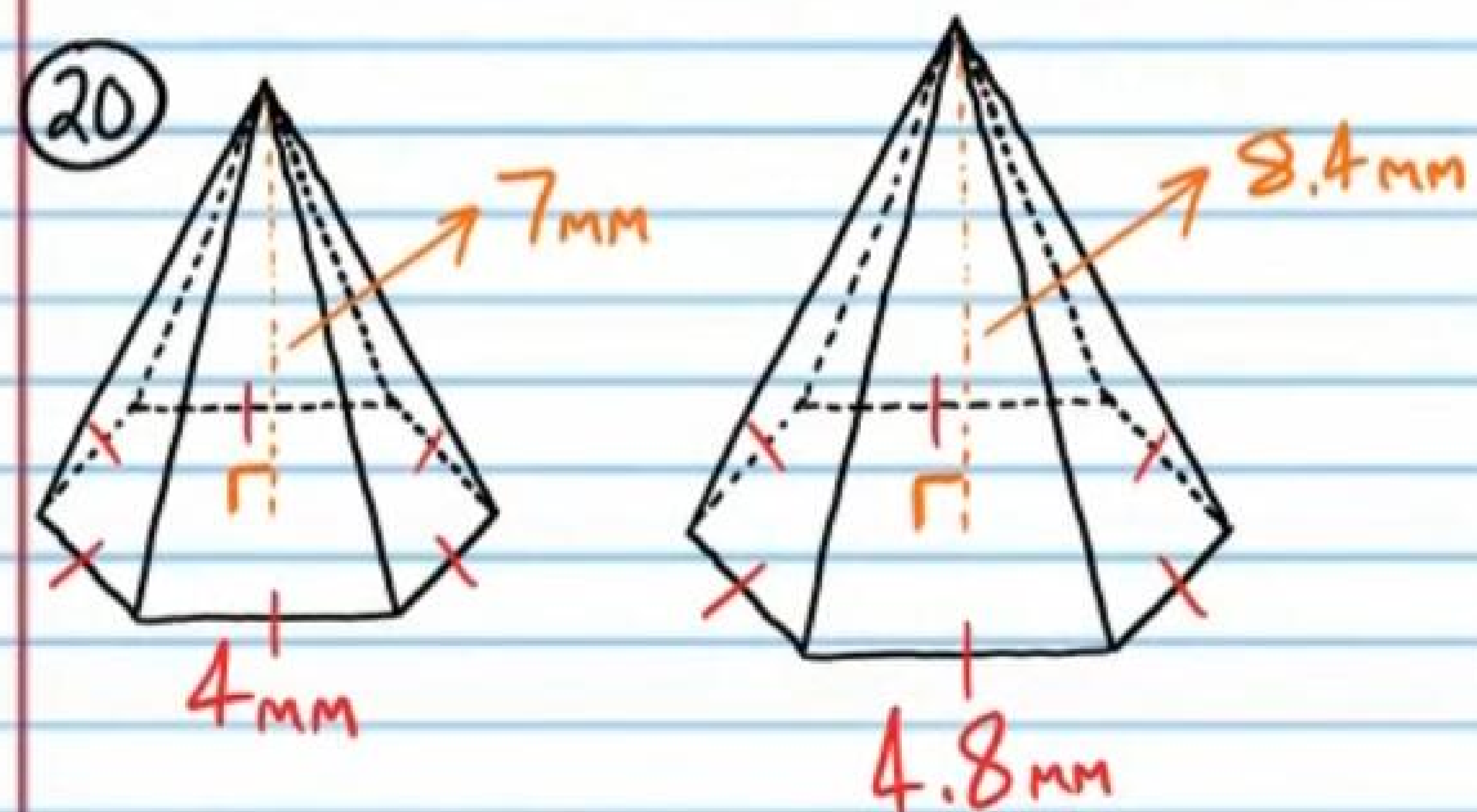
$$V = (1)(3)(5) + \frac{1}{2}(3)(1)(1)$$

$$V = 15 + \frac{3}{2}$$

$$V = \boxed{16.5\ m^3}$$

Determine if the pair of figures below contains similar solids. Write the ratios that prove your answers.

(20)

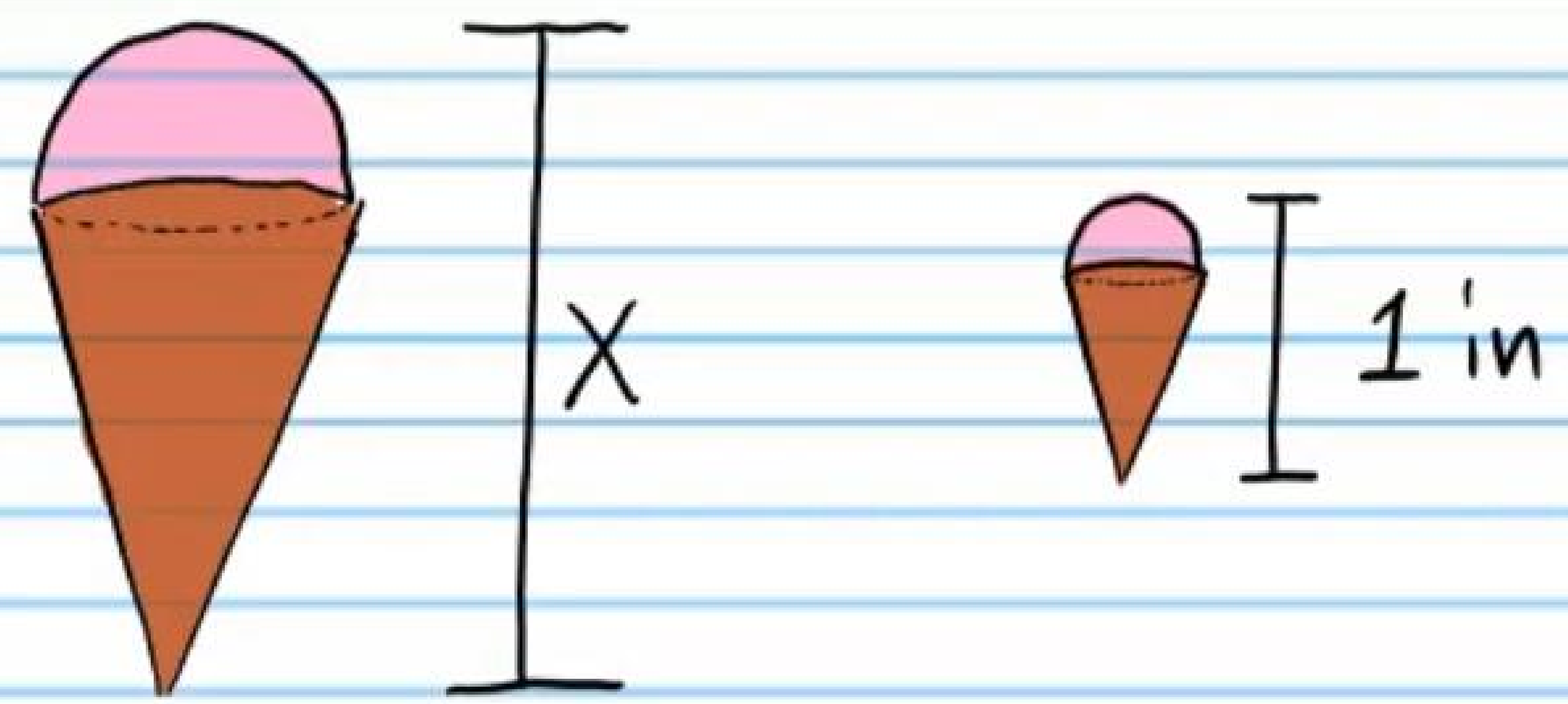


$$\frac{7}{8.4} = \frac{4}{4.8}$$

$$0.8\overline{3} = 0.8\overline{3}$$

yes

(21) If the following solids are similar, find X.



$V_1 = 320 \text{ in}^3$ $\qquad$ $V_2 = 5 \text{ in}^3$

$$\frac{V_1}{V_2} = (SF)^3$$

$$\frac{320}{5} = \left(\frac{X}{1}\right)^3$$

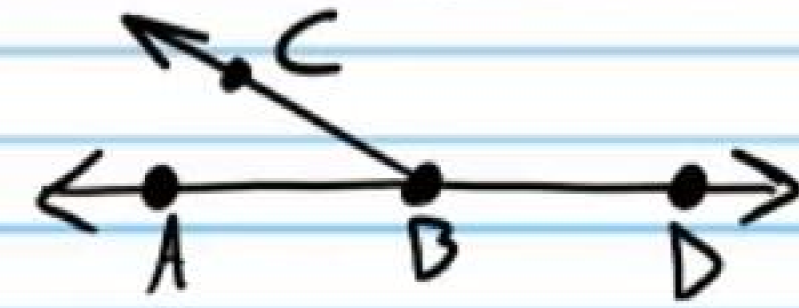
$$64 = x^3$$

$$\sqrt[3]{64} = \sqrt[3]{x^3}$$

$$\boxed{4 \text{ in} = X}$$

Use the two-column proof below to do problem # 22.

If $m\angle ABC = 155°$ in the diagram below, then $m\angle CBD = 25°$.



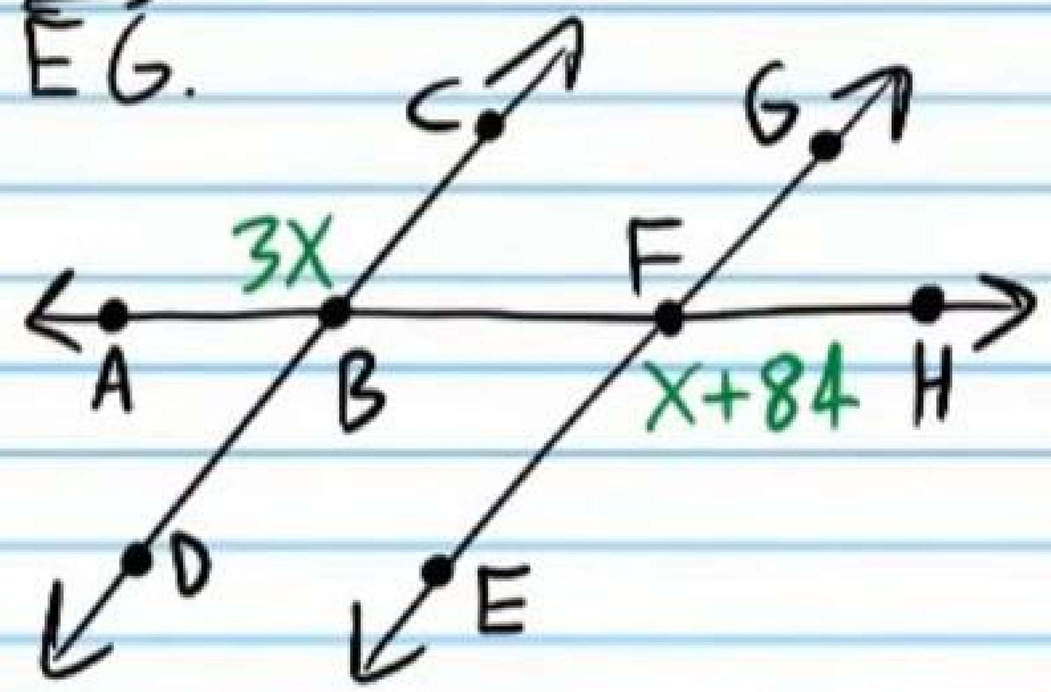
Statement	Reason
I) $\angle ABC$ & $\angle CBD$ form a linear pair.	Given (diagram)
II) $\angle ABC$ & $\angle CBD$ are supplementary angles.	Linear Pair Post.
III) $m\angle ABC + m\angle CBD = 180°$	Def. of supp. $\angle$'s.
IV) $m\angle ABC = 155°$	Given.
V) $155° + m\angle CBD = 180°$	Subs. Prop of =.
VI) $m\angle CBD = 25°$	Subt. Prop of =.

(22) Convert the two-column proof above to paragraph form.

It is given in the diagram that $\angle ABC$ and $\angle CBD$ form a linear pair. Therefore, by the Linear Pair Postulate, they are supplementary angles. By the definition of supplementary angles, $m\angle ABC + m\angle CBD = 180°$. It is also given the $m\angle ABC = 155°$, so by the Substitution Property of Equality, $155° + m\angle CBD = 180°$. Finally, by the Subtraction Property of Equality, $m\angle CBD = 25°$.

Use the statement below to do problem #'s 23-24.

If $x = 42°$, then $\overleftrightarrow{DC} \parallel \overleftrightarrow{EG}$.



(23) Write a flowchart proof for the statement.

$x = 42°$
Given

$m\angle ABC = 3x$ & $m\angle EFH = x+84$
Given (diagram)

↓

$m\angle ABC = 3(42)$ & $m\angle EFH = 42+84$
Subs. Prop of =

↓

$m\angle ABC = 126$ & $m\angle EFH = 126°$
simplification

↓

$m\angle ABC = m\angle EFH$
Trans. Prop of =.

↓

$\angle ABC \cong \angle EFH$
Def. of $\cong$ ∠'s.

→

$\overleftrightarrow{DC} \parallel \overleftrightarrow{EG}$
AEACT

(24) Write a paragraph proof for the statement above.

It is given that $x = 42°$ and the diagram shows us that $m\angle ABC = 3x$ and $m\angle EFH = x+84$. By the Substitution Property of Equality, $m\angle ABC = 3(42)$ and $m\angle EFH = 42+84$. Simplifying, the equations become $m\angle ABC = 126°$ and $m\angle EFH = 126°$. By the Transitive Property of Equality, $m\angle ABC = m\angle EFH$ and therefore by the definition of congruent angles, $\angle ABC \cong \angle EFH$. Finally, we know that $\overleftrightarrow{DC} \parallel \overleftrightarrow{EG}$ by the Alternate Exterior Angles Converse Theorem.

(25) Write a flowchart proof for the following statement: If M is the midpoint of the segment below, and AM = 16, then AB = 32.

M is the midpoint of $\overline{AB}$
Given.

↓

$\overline{AM} \cong \overline{MB}$
Def. of midpoint.

↓

$AM = MB$
Def. of $\cong$ Segs.

$AB = AM + MB$
Seg Add. Post.

↓

$AB = AM + AM$
Subs. Prop of =.

↓

$AB = 2AM$
simplification

$AM = 16$
Given

↓

$AB = 2(16)$
Subs. Prop of =

↓

$AB = 32$
simplification